

ISLAMIC UNIVERSITY OF TECHNOLOGY

EEE4606 Project: Interactive Aircraft Control System Visualizer

Team: Team 00

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1. Objective

The objective of this project is to design and implement an interactive platform that demonstrates how aircraft use feedback control systems to maintain altitude and heading. It helps –

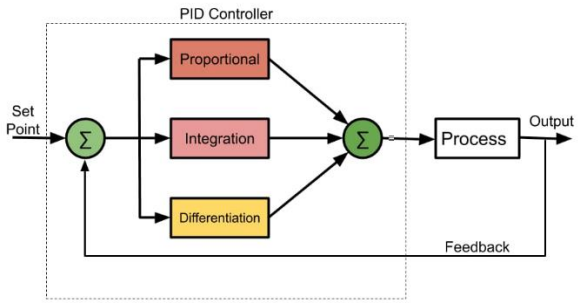
- visualize the effect of disturbances
- the role of PID controllers
- the nested loop structure of aircraft dynamics (pitch–altitude & roll–heading)

2. Introduction

Rather than presenting flight dynamics as abstract equations, our project shows them as visual and interactive behavior. By applying simplified aircraft dynamics with control algorithms, the project allows users to set reference values, introduce disturbances and observe how the system stabilizes itself.

1.1 Terminology

- Pitch: Aircraft nose going up or down
- Elevator: Movable parts of tail (changes pitch)
- Roll: Aircraft tilting left or right
- Aileron: movable parts of wings (changes roll)
- PID Controller:



- PID Controller Effects on System Response:

Parameter	*Rise Time*	*Overshoot*	*Settling Time*	*Steady-State Error*
Kp	Decrease	Increase	Small Change	Decrease
Ki	Decrease	Increase	Increase	Eliminate
Kd	Small Change	Decrease	Decrease	Small Change

3. Working Principle

3.1 How the Project Works

The project models the aircraft as two interconnected control loops:

- Altitude-Pitch Loop: Pitch angle is controlled via elevator deflection; altitude is derived from pitch & forward speed.
- Heading-Roll Loop: Roll angle is controlled via aileron deflection; heading is derived from roll.

User provides a desired altitude and heading. The system compares these references with the current values and calculates the error. PID controllers then adjust elevator and aileron deflections to minimize these errors. If disturbances (step, impulse, sinusoids) are introduced, the controllers work to bring the aircraft back to its reference values. The entire process is visualized in real time, showing how the aircraft dynamically responds.

3.2 System Structure:

- Two nested PID loops:
  - Altitude loop (outer) → Pitch loop (inner)
  - Heading loop (outer) → Roll loop (inner)
- Outer loops (altitude, heading) compute references.
- Inner loops (pitch, roll) drive actual control surfaces.
- Aircraft dynamics + integration give back altitude and heading.

a) Altitude control loop:

Altitude Ref → C_alt (PID) → Pitch Ref → C_pitch (PID) → Elevator → Aircraft → θ → ∫ → Altitude

- Altitude Ref → the altitude wanted (setpoint).
- C_alt (PID) → compares desired altitude vs actual altitude → outputs a pitch reference (how much the nose should go up or down).
- C_pitch (PID) → takes that pitch reference → drives the elevator (the control surface that controls pitch).
- Aircraft → θ → ∫ → Altitude
  - Elevator changes the aircraft's pitch angle (θ).
  - Integrating pitch over time changes the altitude.

b) Heading control loop:

Heading Ref → C_psi (PID) → Roll Ref → C_roll (PID) → Aileron → Aircraft → ϕ → ∫ → Heading

- Heading Ref → the heading wanted (setpoint in degrees).
- C_psi (PID) → compares desired heading vs actual heading → outputs a roll reference (how much the wings should tilt).
- C_roll (PID) → takes that roll reference → drives the aileron (the control surface that controls roll).
- Aircraft → ϕ → ∫ → Heading
  - Aileron changes the roll angle (ϕ).
  - Integrating roll over time changes the heading.

3.3 Function of each PID Controllers:

Four PID controllers are used in a nested-loop structure to regulate altitude and heading through pitch and roll. Each controller plays a different role, but the principle: calculate error (difference between reference and actual value) and apply corrective control through proportional, integral & derivative actions.

a) Altitude PID (Outer Loop)

- Function:
  Compares desired altitude with actual altitude. The output is a pitch reference (i.e., how much the aircraft should pitch up or down).
- Gain Effects:
  - Kp_alt: Higher proportional gain → faster altitude correction but may cause oscillations.
  - Ki_alt: Removes steady-state altitude error (e.g., compensates for persistent offset due to constant disturbance like wind). Too high → slow oscillations.
  - Kd_alt: Reacts to sudden altitude changes (rate of error). Adds stability but excessive value → noise sensitivity.

b) Pitch PID (Inner Loop for Altitude)

- Function:
  Receives the pitch reference from the altitude controller and commands the elevator to achieve that pitch angle.

- Gain Effects:
  - Kp_pitch: Increases responsiveness of pitch to elevator input. Too high → overshoot/instability.
  - Ki_pitch: Corrects pitch bias (e.g., due to trim error). Too high → drift or slow oscillation.
  - Kd_pitch: Smooths out fast pitch oscillations, improves damping. Too high → jitter.

c) Heading PID (Outer Loop)

- Function:
  Compares desired heading with actual heading. The output is a roll reference (how much the aircraft should bank left/right).
- Gain Effects:
  - Kp_psi: Main correction force for heading error. Higher values → quicker heading adjustment. Too high → overshoot.
  - Ki_psi: Corrects for steady heading drift (like crosswind). Excessive values → oscillations around heading.
  - Kd_psi: Stabilizes sudden heading changes. Overuse → sluggishness.

d) Roll PID (Inner Loop for Heading)

- Function:
  Receives the roll reference from the heading controller and commands the aileron to achieve that roll angle.
- Gain Effects:
  - Kp_roll: Makes roll angle follow reference quickly. Too high → roll oscillations.
  - Ki_roll: Corrects long-term roll offset (e.g., imbalance in wings). Too high → instability.
  - Kd_roll: Damps roll oscillations, prevents overshoot. Too high → sluggish control.

4. Code Algorithm

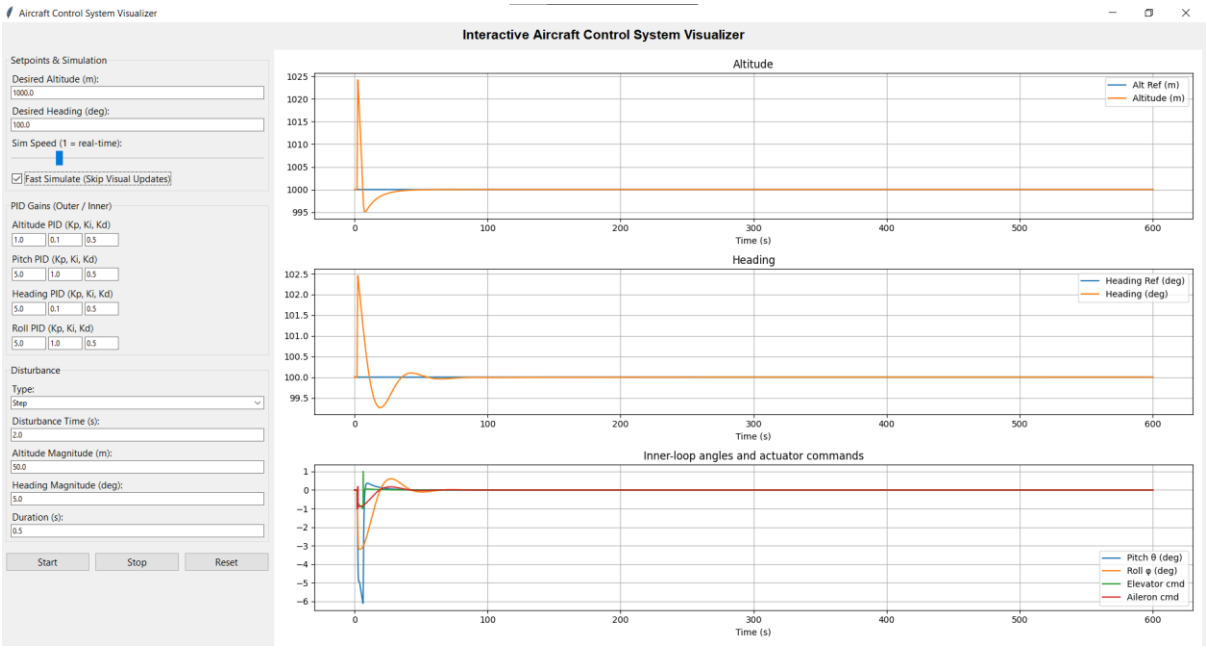
1. **Initialization:**
   - Define parameters: airspeed, pitch/roll gains, natural frequencies, damping ratios, and outer-loop constants for altitude and heading.
2. **Reference Input:**
   - User specifies target altitude and heading.
3. **Error Calculation:**
   - Compute error between reference and current altitude/heading.
4. **PID Controllers:**
   - PID for altitude → generates pitch reference.
   - PID for heading → generates roll reference.
   - Inner PID controllers adjust elevator and aileron to achieve those pitch and roll references.
5. **Aircraft Dynamics Update:**
   - Pitch and roll follow 2nd-order dynamics (defined by k, ω_n, ζ)
   - Altitude rate ≈ $V \cdot \theta$
   - Heading rate ≈ $k_\psi \cdot \varphi$
6. **Disturbance Introduce:**
   - Add step, impulse or sinusoid disturbance signals to test system.
7. **Visualization:**
   - Update & plot altitude, heading, pitch & roll in real time.

5. Features

- **Nested Control Loops:** Two-layer structure (inner loop for attitude, outer loop for altitude/heading), closely mirroring actual autopilot design.
- **User-Defined Setpoints:** Set desired altitude and heading interactively, allowing for flexible testing of control performance.
- **Disturbance Handling:** Supports different types of disturbances (step, impulse, sinusoid), making it possible to evaluate system stability under sudden changes.
- **Dynamic Modeling:** Incorporates natural frequencies (how fast the aircraft reacts) and damping ratios (how smooth the response is) to simulate realistic pitch and roll responses, highlighting the importance of tuning parameters.
- **Interactive Visualization:** Graphical output shows time responses of altitude, heading, pitch, and roll, giving clear insights into how the system reacts and stabilizes.

6. Outcomes:

An Example:



First Figure Explanation:

Shows how aircraft maintains desired altitude with disturbances, with the blue line tracking actual altitude and orange line showing the reference.

Second Figure Explanation:

Shows how aircraft maintains desired heading with disturbances, with the blue line tracking actual altitude and orange line showing the reference.

Third Figure Explanation:

1. Pitch θ (deg) - Blue Line

The actual pitch angle of the aircraft in degrees. Positive pitch means nose up, negative means nose down

2. Roll ϕ (deg) - Orange Line

The actual roll angle of the aircraft in degrees**.** Positive roll means right wing down, negative means left wing down

3. Elevator cmd - Green Line

Elevator command (-1 to +1, where -1 = full down, +1 = full up). Controls pitch angle: negative commands make aircraft pitch down, positive makes pitch up.

4. Aileron cmd - Red Line

Aileron command (-1 to +1, where -1 = left roll, +1 = right roll). Controls roll angle for heading changes

Observations:

- Elevator leads, pitch follows: The elevator command happens first, then pitch angle changes
- Aileron leads, roll follows: The aileron command happens first, then roll angle changes